

ACADEMIC DETAILS

Degree/Certificate	Specialization	Institute/Board	Year	CPI/%
B.Tech.	Mechanical Engineering	Indian Institute of Technology Gandhinagar	2024–2028	8.70
Senior Secondary	Physics, Chemistry, Mathematics	Late P. B. Jog Junior College (MSBSHSE)	2024	89.5
Secondary	SSC	Mahesh Vidyalaya English Medium School	2022	95.8

PROJECTS

- **Neural Network Framework for Learning Constitutive Laws in 2D Granular Flow**

[Jan 2026 – Apr 2026]

(Prof. Harkirat Singh, IIT Gandhinagar) | [Project Link](#)

- **Engineered** a modular PyTorch deep learning framework, evaluating **3** critical internal stress components (σ_{xx} , σ_{yy} , σ_{xy}) across **25** hyperparameter configurations to accurately map nonlinear physics paths.
 - **Optimized** overall cross-dataset prediction error rates from **48%** down to **3%** by implementing math-based reference normalization models, confirming robust generalization on **2** unseen datasets.
 - **Streamlined** the data preprocessing and training pipeline by automating 2D spatial grid mapping routines to handle large-scale internal 2D stress tensor variations.
- **SmartSpend AI – Full Stack Personal Finance Tracker with AI Budget Coach**

[May 2026 – Jun 2026]

[Project Link](#) | [Live Demo](#)

- **Built** a responsive full-stack finance web application, executing automated tracking logic across **100+** personal entries to precisely evaluate custom categorical budgetary allocation thresholds.
 - **Streamlined** personalized budgeting advice across **5** financial categories by integrating the conversational Google Gemini API with a robust React, Node.js, Express, and MongoDB backend architecture.
 - **Optimized** database sorting lookups and data monitoring workflows by structuring backend tracking across **3** distinct data schemas, engineering secure user authentication and interactive interface dashboard layouts.
- **Validation of Angular Momentum Conservation via Reynolds Transport Theorem**

[Jan 2026 – Apr 2026]

(Prof. Shubhangi Bansude, IIT Gandhinagar) | [Project Link](#)

- **Validated** physical angular momentum verification milestones by fabricating a custom **3-nozzle** rotating water sprinkler mechanism prototype from scratch following formal mathematical validation rules.
 - **Modelled** relative error variances of **31.1%** and **42.0%** across **2** distinct operating speeds by formulating a structural fluid predictive framework balancing exit velocity and rotational speed metrics.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, PyTorch (Basic)
- **Developer Tools:** Git, GitHub, Visual Studio Code
- **Software & Simulation Utilities:** MATLAB, Cantera, Mathematica, Autodesk Inventor, MSC Adams (Basic), ANSYS Fluent (Basic), Creality Print

RELEVANT COURSEWORK

- **Computer Science & Mathematics:** Data Structures and Algorithms, Probability, Statistics and Data Visualization, Calculus of Single Variable and Linear Algebra

ACHIEVEMENTS

- **Awarded the Dean’s List** (top 5% of class) and secured **Rank 1 (Branch Topper)** in Mechanical Engineering with an **SPI of 9.48** in Semester 3.
- **Awarded an A+ (11/10)** grade in **Thermodynamics** for exceptional academic performance and accuracy across advanced curriculum evaluation structures.
- **Contributed** as a team member of the **8-member** IIT Gandhinagar squad to secure **First Position** out of **250+** teams in Division 4 of the India College Chess Championship 2025, claiming a **INR 60,000** prize.
- **Secured Team Runner-up** honors at the DA-IICT Chess Tournament and individual **Runner-up (Unrated Category)** recognition at the competitive IIT Bombay Sports Fest.

POSITIONS OF RESPONSIBILITY

- **Coordinating Team Member, Institute Sports Events, IIT Gandhinagar**

[Jul 2024 – Apr 2026]

- **Managed** basic operational logistics, on-ground scheduling configurations, and formal tracker loops for sports festivals while collaborating closely with student council representatives to facilitate tournament timelines.
- **Junior Team Member, Amalthea & Blithchron Festivals, IIT Gandhinagar**

[Aug 2024 – Feb 2025]

- **Coordinated** core event operations during Amalthea’25 by assisting with hospitality logistics and tracked company outreach alongside record matrices to maintain partner communications for Blithchron’25.

EXTRA-CURRICULAR ACTIVITIES

- **Represented** IIT Gandhinagar twice consecutively at the national **Inter IIT Sports Meet**, competing in Chess matches against varsity chapters from **over 20** participating IITs.
- **Participated** actively in campus tournament brackets and regional inter-university chess invitations hosted at external venues including DA-IICT, Ahmedabad University, and IIT Jodhpur.